

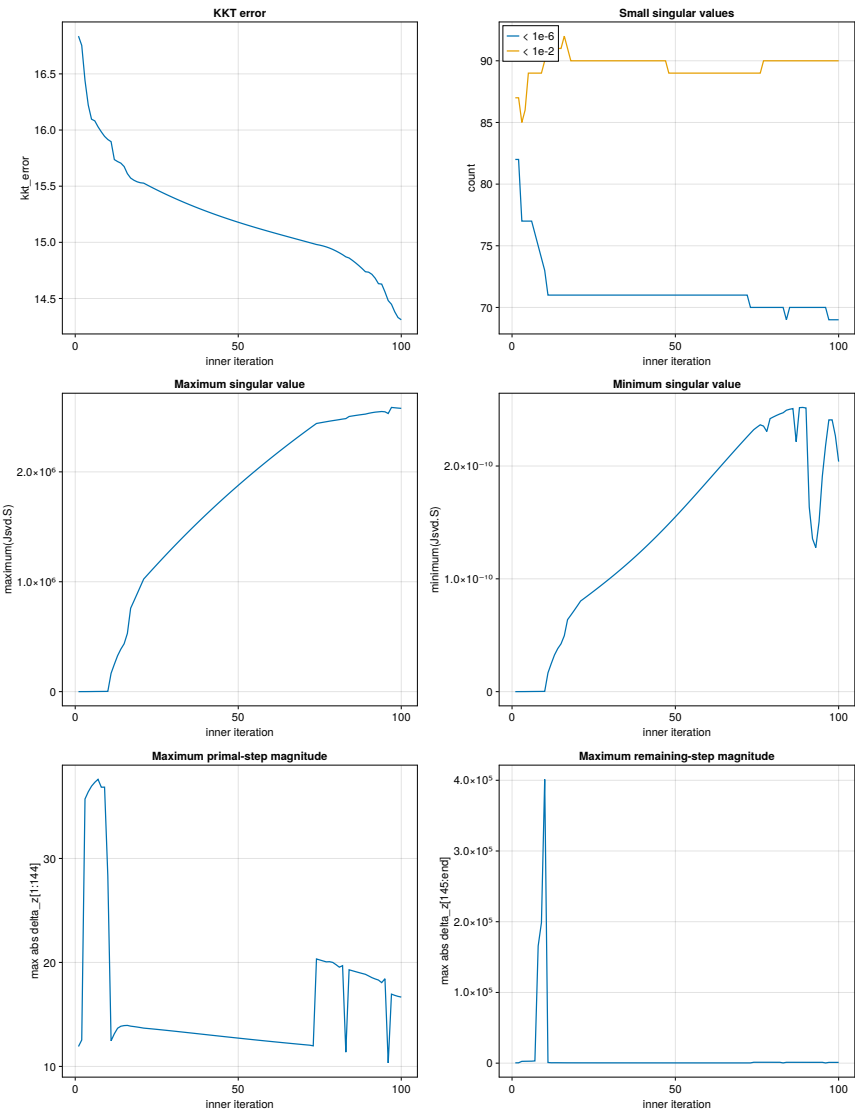
Solver Diagnostics: Naive Solve

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
kkt_error	1.431040e+01	1.683624e+01	1.523837e+01	1.517419e+01
count(Jsvd.S < 1e-6)	6.900000e+01	8.200000e+01	7.127000e+01	7.100000e+01
count(Jsvd.S < 1e-2)	8.500000e+01	9.200000e+01	8.959000e+01	9.000000e+01
maximum(Jsvd.S)	4.591747e+00	2.586359e+06	1.672943e+06	1.890061e+06
minimum(Jsvd.S)	1.118702e-16	2.519356e-10	1.441471e-10	1.496996e-10
maximum(abs.(delta_z[1:144]))	1.033239e+01	3.760888e+01	1.620384e+01	1.341753e+01
maximum(abs.(delta_z[145:end]))	2.942354e+02	4.013592e+05	8.374970e+03	4.459977e+02

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\max \delta z_{145:} $
1	1.683624e+01	82	87	4.591747e+00	1.118702e-16	1.191653e+01	4.631239e+02
2	1.675363e+01	82	87	2.770448e+02	2.524619e-14	1.254051e+01	4.980500e+02
3	1.644180e+01	77	85	3.874278e+02	3.790864e-14	3.571468e+01	2.558139e+03
4	1.622232e+01	77	86	5.704977e+02	5.573447e-14	3.640686e+01	2.662246e+03
5	1.609664e+01	77	89	9.230953e+02	8.953808e-14	3.694644e+01	2.747190e+03
6	1.608168e+01	77	89	1.388056e+03	1.276213e-13	3.731702e+01	2.819993e+03
7	1.602873e+01	76	89	1.622255e+03	1.532354e-13	3.760888e+01	3.145670e+03
8	1.598391e+01	75	89	1.859900e+03	1.779991e-13	3.684954e+01	1.656827e+05
9	1.594377e+01	74	89	2.080718e+03	2.037223e-13	3.685560e+01	1.986098e+05
10	1.591552e+01	73	90	2.303025e+03	2.213712e-13	2.830815e+01	4.013592e+05
11	1.589653e+01	71	91	1.668563e+05	1.662741e-11	1.246200e+01	9.298524e+02
12	1.573694e+01	71	91	2.489920e+05	2.481852e-11	1.313700e+01	5.633083e+02
13	1.571912e+01	71	91	3.273734e+05	3.252868e-11	1.367048e+01	5.893743e+02
14	1.570511e+01	71	91	3.870762e+05	3.818587e-11	1.386396e+01	5.824525e+02
15	1.567500e+01	71	91	4.354634e+05	4.240871e-11	1.392437e+01	5.476872e+02
16	1.561165e+01	71	92	5.299657e+05	4.957489e-11	1.394246e+01	5.122515e+02
17	1.557234e+01	71	91	7.589994e+05	6.380792e-11	1.387052e+01	4.845608e+02
18	1.555289e+01	71	90	8.244257e+05	6.778291e-11	1.383072e+01	4.777292e+02
19	1.553892e+01	71	90	8.909296e+05	7.187503e-11	1.378579e+01	4.711626e+02
20	1.553038e+01	71	90	9.577873e+05	7.606337e-11	1.373611e+01	4.648624e+02
21	1.552710e+01	71	90	1.024476e+06	8.033680e-11	1.368221e+01	4.588217e+02
22	1.551170e+01	71	90	1.057352e+06	8.248435e-11	1.365349e+01	4.558668e+02
23	1.549661e+01	71	90	1.090049e+06	8.465103e-11	1.362397e+01	4.529727e+02
24	1.548181e+01	71	90	1.122541e+06	8.683775e-11	1.359373e+01	4.501375e+02
25	1.546729e+01	71	90	1.154810e+06	8.904570e-11	1.356285e+01	4.473593e+02
26	1.545305e+01	71	90	1.186840e+06	9.127624e-11	1.353138e+01	4.446361e+02
27	1.543907e+01	71	90	1.218620e+06	9.353091e-11	1.349939e+01	4.419660e+02
28	1.542536e+01	71	90	1.250141e+06	9.581133e-11	1.346695e+01	4.393471e+02
29	1.541191e+01	71	90	1.281396e+06	9.811919e-11	1.343412e+01	4.367773e+02
30	1.539870e+01	71	90	1.312382e+06	1.004562e-10	1.340094e+01	4.342549e+02
31	1.538574e+01	71	90	1.343096e+06	1.028240e-10	1.336746e+01	4.317779e+02
32	1.537301e+01	71	90	1.373537e+06	1.052242e-10	1.333375e+01	4.293445e+02
33	1.536052e+01	71	90	1.403705e+06	1.076584e-10	1.329982e+01	4.269532e+02
34	1.534825e+01	71	90	1.433600e+06	1.101279e-10	1.326574e+01	4.246020e+02
35	1.533621e+01	71	90	1.463225e+06	1.126340e-10	1.323153e+01	4.222896e+02
36	1.532438e+01	71	90	1.492581e+06	1.151779e-10	1.319723e+01	4.200142e+02
37	1.531277e+01	71	90	1.521671e+06	1.177605e-10	1.316286e+01	4.177745e+02
38	1.530136e+01	71	90	1.550499e+06	1.203825e-10	1.312847e+01	4.155690e+02
39	1.529015e+01	71	90	1.579066e+06	1.230446e-10	1.309407e+01	4.133964e+02
40	1.527914e+01	71	90	1.607378e+06	1.257471e-10	1.305968e+01	4.112554e+02
41	1.526833e+01	71	90	1.635437e+06	1.284900e-10	1.302533e+01	4.091447e+02
42	1.525770e+01	71	90	1.663247e+06	1.312735e-10	1.299104e+01	4.070631e+02
43	1.524725e+01	71	90	1.690811e+06	1.340971e-10	1.295683e+01	4.050096e+02
44	1.523698e+01	71	90	1.718134e+06	1.369604e-10	1.292271e+01	4.029831e+02
45	1.522688e+01	71	90	1.745219e+06	1.398627e-10	1.288869e+01	4.009825e+02
46	1.521695e+01	71	90	1.772069e+06	1.428030e-10	1.285480e+01	3.990069e+02
47	1.520718e+01	71	90	1.798688e+06	1.457803e-10	1.282104e+01	3.970553e+02
48	1.519757e+01	71	89	1.825081e+06	1.487932e-10	1.278742e+01	3.951270e+02
49	1.518810e+01	71	89	1.851249e+06	1.518402e-10	1.275395e+01	3.932209e+02
50	1.517878e+01	71	89	1.877196e+06	1.549198e-10	1.272065e+01	3.913365e+02
51	1.516960e+01	71	89	1.902926e+06	1.580299e-10	1.268751e+01	3.894727e+02
52	1.516055e+01	71	89	1.928442e+06	1.611687e-10	1.265455e+01	3.876291e+02
53	1.515162e+01	71	89	1.953746e+06	1.643339e-10	1.262177e+01	3.858048e+02
54	1.514281e+01	71	89	1.978842e+06	1.675234e-10	1.258918e+01	3.839992e+02
55	1.513412e+01	71	89	2.003733e+06	1.707348e-10	1.255678e+01	3.822116e+02
56	1.512553e+01	71	89	2.028420e+06	1.739655e-10	1.252458e+01	3.804416e+02
57	1.511704e+01	71	89	2.052907e+06	1.772129e-10	1.249257e+01	3.786884e+02
58	1.510864e+01	71	89	2.077196e+06	1.804744e-10	1.246077e+01	3.769516e+02
59	1.510033e+01	71	89	2.101289e+06	1.837472e-10	1.242917e+01	3.752307e+02
60	1.509209e+01	71	89	2.125189e+06	1.870286e-10	1.239778e+01	3.735250e+02
61	1.508393e+01	71	89	2.148898e+06	1.903156e-10	1.236659e+01	3.718343e+02
62	1.507583e+01	71	89	2.172419e+06	1.936054e-10	1.233562e+01	3.701580e+02

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\max \delta z_{145:} $
63	1.506778e+01	71	89	2.195751e+06	1.968951e-10	1.230485e+01	3.684956e+02
64	1.505978e+01	71	89	2.218899e+06	2.001817e-10	1.227429e+01	3.668469e+02
65	1.505182e+01	71	89	2.241864e+06	2.034622e-10	1.224394e+01	3.652113e+02
66	1.504389e+01	71	89	2.264647e+06	2.067337e-10	1.221380e+01	3.635886e+02
67	1.503598e+01	71	89	2.287250e+06	2.099932e-10	1.218387e+01	3.619783e+02
68	1.502809e+01	71	89	2.309674e+06	2.132379e-10	1.215414e+01	3.603802e+02
69	1.502021e+01	71	89	2.331922e+06	2.164647e-10	1.212462e+01	3.587939e+02
70	1.501232e+01	71	89	2.353994e+06	2.196707e-10	1.209530e+01	3.572192e+02
71	1.500443e+01	71	89	2.375893e+06	2.228530e-10	1.206618e+01	3.556557e+02
72	1.499651e+01	71	89	2.397618e+06	2.260089e-10	1.203727e+01	3.541033e+02
73	1.498857e+01	70	89	2.419172e+06	2.291353e-10	1.197714e+01	3.565528e+02
74	1.498051e+01	70	89	2.440454e+06	2.322170e-10	2.033314e+01	1.258087e+03
75	1.497495e+01	70	89	2.445749e+06	2.344359e-10	2.023722e+01	1.252017e+03
76	1.496788e+01	70	89	2.451043e+06	2.365333e-10	2.014378e+01	1.246239e+03
77	1.495922e+01	70	90	2.456295e+06	2.353835e-10	2.005319e+01	1.240782e+03
78	1.494893e+01	70	90	2.461469e+06	2.305853e-10	2.006986e+01	1.246652e+03
79	1.493666e+01	70	90	2.466199e+06	2.419930e-10	1.999796e+01	1.242392e+03
80	1.492261e+01	70	90	2.470737e+06	2.434984e-10	1.978286e+01	1.224721e+03
81	1.490709e+01	70	90	2.475483e+06	2.448922e-10	1.954383e+01	1.205641e+03
82	1.489015e+01	70	90	2.480481e+06	2.461737e-10	1.968440e+01	1.222040e+03
83	1.487081e+01	70	90	2.484614e+06	2.472202e-10	1.136930e+01	3.464843e+02
84	1.486032e+01	69	90	2.503873e+06	2.494543e-10	1.929160e+01	1.193876e+03
85	1.483957e+01	70	90	2.508742e+06	2.503234e-10	1.920038e+01	1.188761e+03
86	1.481700e+01	70	90	2.513373e+06	2.509138e-10	1.911115e+01	1.183876e+03
87	1.479259e+01	70	90	2.517744e+06	2.215081e-10	1.902404e+01	1.179229e+03
88	1.476633e+01	70	90	2.521836e+06	2.517499e-10	1.893920e+01	1.174832e+03
89	1.473821e+01	70	90	2.525627e+06	2.519356e-10	1.885682e+01	1.170700e+03
90	1.473411e+01	70	90	2.532871e+06	2.514813e-10	1.869968e+01	1.163490e+03
91	1.471502e+01	70	90	2.538719e+06	1.635754e-10	1.853637e+01	1.152947e+03
92	1.468049e+01	70	90	2.543399e+06	1.354856e-10	1.841073e+01	1.151813e+03
93	1.463199e+01	70	90	2.546285e+06	1.276803e-10	1.830346e+01	1.149420e+03
94	1.462795e+01	70	90	2.549213e+06	1.506060e-10	1.805594e+01	1.140084e+03
95	1.455892e+01	70	90	2.546650e+06	1.908658e-10	1.843115e+01	1.191643e+03
96	1.448108e+01	70	90	2.531222e+06	2.182219e-10	1.033239e+01	2.942354e+02
97	1.444825e+01	69	90	2.586359e+06	2.408800e-10	1.694893e+01	1.080733e+03
98	1.438145e+01	69	90	2.583338e+06	2.409393e-10	1.683350e+01	1.077088e+03
99	1.433052e+01	69	90	2.580444e+06	2.267330e-10	1.673590e+01	1.074721e+03
100	1.431040e+01	69	90	2.578001e+06	2.039329e-10	1.666559e+01	1.074306e+03